SOLOBOLO / RIEMANN-HYPOTHESIS-FINAL

Hilbert-Vector Resolvents and the Riemann Hypothesis

A detailed analytic seed criterion

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Abstract

We give a precise, self-contained analytic formulation of a Hilbert-vector anchor for the Riemann hypothesis. Let Θ be the entire function determined by $\Theta(w^2) = \xi(\frac{1}{2} + w)$, and put $L(a) = \Theta'(a)/\Theta(a)$ for $a \geq 0$. We prove that RH is equivalent to the existence of one Hilbert space, one bounded nonnegative self-adjoint operator K, and one fixed vector v such that $\frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|(I + aK)^{-1}v\|^2$ on a nonempty positive interval. The derivation includes the resolvent calculus, its divided-difference Gram kernel, and the analytic continuation that excludes zeros off the negative real axis. A possible C/a ambiguity is removed using $\Theta(0) \neq 0$, rather than suppressed by an unstated normalization. The reverse implication constructs a positive diagonal operator from the zero product under RH, with multiplicities counted correctly. We also develop the spectral-measure and all-order moment formulations, explain basepoint shifts, and prove that no exactly calibrated finite-dimensional model can suffice. The result is an analytic equivalence; it does not construct the required seed unconditionally or produce an accepted finite witness.

Evidence and scope. Every proof in this note is an on-paper derivation. No Lean compiler or axiom audit was run. The seed predicates below are defined explicitly here; they are not asserted to be recovered field-for-field definitions of a legacy `HasSeed` declaration. The repository's finite-checker target remains open. Classical background is cited, and no novelty or unconditional proof of RH is claimed.

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1 The exact assertion and its role

The starting point is the family of identities

$$R_a = (I + aK)^{-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|R_a v\|^2, \qquad V_a = 2\sqrt{a}\, R_a v. \tag{1.1}$$

Their significance depends on their quantifiers. The operator and vector must be the *same* for every parameter in the interval, and the scalar L must come from the actual completed Riemann zeta function. An arbitrary nonnegative scalar function can be represented as a squared norm by allowing its vector to vary freely. Such a pointwise representation contains none of the resolvent compatibility used below.

We fix the standard completion

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1.2}$$

with the removable values understood. The symmetry $\xi(s) = \xi(1-s)$ and normalization are classical [1]. Section 2 defines Θ without choosing a square-root branch and proves that L is a well-defined real analytic function on $[0, \infty)$.

Definition 1.1 (Calibrated Hilbert seed)**.** The assertion HSeed_Θ means that there exist a complex Hilbert space $\mathcal{H}$, an operator $K \in \mathcal{B}(\mathcal{H})$, a vector $v \in \mathcal{H}$, and real numbers $0 < \alpha < \beta < \infty$ such that

$$K = K^*, \qquad \langle x, Kx\rangle \geq 0 \quad (x \in \mathcal{H}), \tag{1.3}$$

and, for every $a \in (\alpha, \beta)$,

$$L(a) = \langle v, (I + aK)^{-1} v\rangle. \tag{1.4}$$

The inner product is conjugate-linear in its first slot and linear in its second slot. In particular, the complex extension $z \mapsto \langle v, (I + zK)^{-1}v\rangle$ is holomorphic where the inverse exists.

Definition 1.2 (Differential Hilbert seed)**.** The assertion DSeed_Θ has the same existential data and operator hypotheses as Definition 1.1, but replaces (1.4) by

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a)) = \|(I + aK)^{-1}v\|^2 \quad \text{for every } a \in (\alpha, \beta). \tag{1.5}$$

This is an exact functional identity on an interval, not a test at finitely many sample points.

Theorem 1.3 (Main analytic criterion)**.** *For the actual Θ associated with (1.2),*

$$\mathrm{RH} \iff \mathsf{HSeed}_\Theta \iff \mathsf{DSeed}_\Theta. \tag{1.6}$$

If either seed identity holds on one nonempty positive interval, its calibration extends to every $a \geq 0$ for the same K and v. Under RH, a seed can be chosen on $\ell^2(\mathbb{N}; \mathbb{C})$ with K positive, compact and trace class. In that realization,

$$\operatorname{Tr} K = \|v\|^2 = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{1.7}$$

Sections 3–6 prove the forward and differential implications without assuming RH or a positive zero product. Section 7 proves the converse under RH. The theorem is a criterion: the existential seed assertion carries the mathematical difficulty of RH. No independently constructed seed is supplied here.

There are two different questions behind this note. The analytic question asks exactly what a fixed positive resolvent identity implies. The finite question asks whether a terminating checker on finite raw data can certify such an identity, and whether RH guarantees an accepted input to that same checker. The second question does not follow merely by naming the first one a seed. Section 10 makes this distinction precise.

2 The actual entire function and its zero geometry

2.1 Square descent without a branch choice

Put $E(w) = \xi(\frac{1}{2} + w)$. This is entire, and the functional equation makes it even. Its Taylor series has the form

$$E(w) = \sum_{n=0}^{\infty} c_n w^{2n}, \qquad c_n = \frac{\xi^{(2n)}(1/2)}{(2n)!}. \tag{2.1}$$

Define

$$\Theta(z) = \sum_{n=0}^{\infty} c_n z^n. \tag{2.2}$$

The series has infinite radius of convergence. Indeed, for any $r > 0$, the absolute convergence of (2.1) at a point with $|w| = \sqrt{r}$ gives $\sum_n |c_n| r^n < \infty$. Consequently

$$\Theta(w^2) = \xi(\tfrac{1}{2} + w) \qquad (w \in \mathbb{C}). \tag{2.3}$$

The shorthand $\Theta(z) = \xi(\frac{1}{2} + \sqrt{z})$ therefore has a single-valued meaning: the two square roots give the same value. The branch cut introduced later belongs to the resolvent domain, not to Θ.

Complex conjugation preserves ξ, as follows first on its defining real domain and then by analytic continuation. Thus the coefficients c_n are real, and $\Theta(\overline{z}) = \overline{\Theta(z)}$.

2.2 Positivity on the real positive axis

For completeness, the nonvanishing needed to define L can be checked without RH. For $0 < s < 1$, the alternating-series formula [2, Eq. 25.2.3] gives

$$\zeta(s) = \frac{\eta(s)}{1 - 2^{1-s}}, \qquad \eta(s) = \sum_{n=1}^{\infty} (-1)^{n-1} n^{-s} > 0. \tag{2.4}$$

The strict inequality follows by grouping consecutive terms: $(2j-1)^{-s} - (2j)^{-s} > 0$. The even partial sums converge to the same alternating-series limit and already their first pair is positive. The denominator in (2.4) is negative, so $\zeta(s) < 0$. Since $s(s-1) < 0$ and the remaining factors in (1.2) are positive, $\xi(s) > 0$ on $(0, 1)$. For $s > 1$, every factor in (1.2) is positive, and the removable value at $s = 1$ is $1/2$. Applying this with $s = 1/2 + \sqrt{a}$ proves

$$\Theta(a) > 0 \quad (a \geq 0), \qquad \Theta(0) = \xi(1/2) > 0. \tag{2.5}$$

It follows that

$$L(a) = \frac{\Theta'(a)}{\Theta(a)} \in \mathbb{R} \qquad (a \geq 0) \tag{2.6}$$

is real analytic, including at zero. On this axis it equals the recovered quantity $\mathrm{Re}(\Theta'(a)/\Theta(a))$; taking the real part loses no information there. It would not be legitimate to make that replacement for arbitrary complex z.

Expanding at zero gives the exact normalization

$$\kappa := L(0) = \frac{\Theta'(0)}{\Theta(0)} = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{2.7}$$

No numerical approximation to κ is needed in any proof below.

2.3 The critical line becomes the negative real axis

The zeros of ξ are exactly the nontrivial zeros of ζ; the standard zero symmetry and the meaning of RH are recalled in [3]. If $\rho = 1/2 + w$ is such a zero, then $z = w^2$ is a zero of Θ, and conversely every zero of Θ arises this way by (2.3).

For $w = x + iy$,

$$w^2 = (x^2 - y^2) + 2ixy. \tag{2.8}$$

If w^2 is real and nonpositive, then $xy = 0$ and $x^2 - y^2 \le 0$. If $y = 0$, these imply $x = 0$; if $y \neq 0$, the equality $xy = 0$ again implies $x = 0$. Conversely $x = 0$ gives $w^2 = -y^2 \le 0$. Since $\Theta(0) \neq 0$, its zeros never correspond to $w = 0$. We have therefore proved

Proposition 2.1 (Exact zero geometry)**.**

$$\mathrm{RH} \quad \Longleftrightarrow \quad \big(\Theta(z) = 0 \implies z \in (-\infty, 0)\big). \tag{2.9}$$

This elementary change of variables is the final geometric step of the forward proof. It refers to the actual ξ and Θ throughout.

3 Resolvent calculus with one fixed vector

Fix a complex Hilbert space $\mathcal{H}$, a bounded nonnegative self-adjoint operator K, and a vector v. For $a \ge 0$, set

$$R_a = (I + aK)^{-1}, \qquad f(a) = \langle v, R_a v\rangle, \qquad g(a) = af(a). \tag{3.1}$$

At this stage f is a model scalar function; equality with the actual L has not been assumed.

3.1 Invertibility, self-adjointness and commutation

For every $x \in \mathcal{H}$, positivity and Cauchy–Schwarz imply

$$\|x\|^2 \le \langle x, (I + aK)x\rangle \le \|x\|\,\|(I + aK)x\|.$$

Thus $I + aK$ is bounded below by 1. Its range is closed. Its orthogonal complement is the kernel of its adjoint, which equals its own kernel because the operator is self-adjoint. The lower bound makes that kernel zero, so the range is also dense and hence is all of $\mathcal{H}$. It follows that

$$\|R_a\| \le 1, \qquad R_a^* = R_a, \qquad R_a \ge 0. \tag{3.2}$$

For the last inequality write $y = (I + aK)x$ and observe that $\langle y, R_a y\rangle = \langle (I + aK)x, x\rangle \ge 0$. The operator K commutes with $I + aK$ and therefore with its inverse. The operators R_a and R_b also commute.

Lemma 3.1 (Two resolvent identities)**.** *For $a, b \ge 0$,*

$$R_a - R_b = (b - a)R_a K R_b, \tag{3.3}$$

$$aR_a - bR_b = (a - b)R_a R_b. \tag{3.4}$$

Proof. For the first identity, multiply out $R_a[(I + bK) - (I + aK)]R_b$. For the second, the same calculation with $a(I + bK) - b(I + aK) = (a - b)I$ gives the result. Both formulas also hold when $a = b$, with both sides then zero. □

3.2 Differentiation in operator norm

From (3.3),

$$\frac{R_{a+h} - R_a}{h} = -R_{a+h} K R_a.$$

Continuity of inversion, or the same identity and local boundedness, shows that the right side tends in operator norm to $-R_a K R_a$. Since K commutes with R_a,

$$R_a' = -K R_a^2. \tag{3.5}$$

Repeated differentiation yields

$$R_a^{(n)} = (-1)^n n! K^n R_a^{n+1} \qquad (n \geq 0). \tag{3.6}$$

One can verify the induction directly: differentiating R_a^{n+1} produces $n+1$ identical commuting terms $-K R_a^{n+2}$. Applying the continuous linear functional $T \mapsto \langle v, Tv \rangle$ gives

$$(-1)^n f^{(n)}(a) = n! \langle v, K^n R_a^{n+1} v \rangle \geq 0. \tag{3.7}$$

The last operator is positive, for example by the bounded self-adjoint functional calculus [6]. Thus f is completely monotone. This consequence alone will not characterize a calibrated seed.

3.3 The norm-square identity

The identity $(I + aK)R_a = I$ implies $R_a = R_a^2 + aK R_a^2$. Consequently

$$\begin{aligned} g'(a) &= f(a) + a f'(a) \\ &= \langle v, (R_a - aK R_a^2) v \rangle \\ &= \langle v, R_a^2 v \rangle = \|R_a v\|^2. \end{aligned} \tag{3.8}$$

This proves the central algebraic formula, including its exact coefficient. It involves $af(a)$, not $f(a)$ alone and not an unspecified change of basepoint. Differentiating further gives

$$(-1)^n g^{(n+1)}(a) = (n+1)! \langle v, K^n R_a^{n+2} v \rangle \geq 0 \qquad (n \geq 0). \tag{3.9}$$

In particular g is increasing and concave on the positive axis. If $v \neq 0$, then $R_a v \neq 0$ and $g'(a) > 0$.

4 The Gram kernel and its exact normalization

Define $V_a = 2\sqrt{a}\, R_a v$ for $a > 0$. Taking the scalar product of (3.4) with v and using self-adjointness gives, for $a \neq b$,

$$\frac{g(a) - g(b)}{a - b} = \langle v, R_a R_b v \rangle = \langle R_a v, R_b v \rangle. \tag{4.1}$$

Hence

$$\langle V_a, V_b \rangle = 4\sqrt{ab}\, \frac{a f(a) - b f(b)}{a - b} \qquad (a, b > 0,\ a \neq b). \tag{4.2}$$

If the calibration $f = L$ holds, this is precisely the recovered Hilbert-vector Gram identity.

Proposition 4.1 (Diagonal and positivity)**.** *The divided-difference kernel*

$$\mathcal{K}_g(a,b) = \begin{cases} \dfrac{g(a)-g(b)}{a-b}, & a \neq b, \\ g'(a), & a = b \end{cases} \tag{4.3}$$

is positive semidefinite. More precisely, for any finite collection $a_1, \dots, a_m > 0$ and complex numbers $c_1, \dots, c_m$,

$$\sum_{i,j=1}^{m} \overline{c_i} c_j \mathcal{K}_g(a_i, a_j) = \|\sum_{j=1}^{m} c_j R_{a_j} v\|^2 \geq 0. \tag{4.4}$$

The diagonal extension of (4.2) is

$$\|V_a\|^2 = 4a\, g'(a) = 4a\, \|R_a v\|^2. \tag{4.5}$$

Proof. Equation (4.1) gives all off-diagonal entries, while (3.8) gives the diagonal. Expanding the displayed squared norm proves (4.4); multiplication by the factors $2\sqrt{a_i}$ gives the scaled Gram matrix. □

For two distinct parameters, the determinant condition is the concrete inequality

$$\left|\frac{g(a)-g(b)}{a-b}\right|^2 \leq g'(a)g'(b). \tag{4.6}$$

It is a necessary condition tying different points together. Positivity of g' at every individual point does not imply it.

Example 4.2 (Even strong scalar positivity is insufficient)**.** Let $\widetilde{g}(a) = 1 - e^{-a}$ and $\widetilde{f}(a) = (1-e^{-a})/a$ for $a > 0$. Then $\widetilde{g}' > 0$, $\widetilde{g}'' < 0$, and

$$\widetilde{f}(a) = \int_0^1 e^{-at}\,\mathrm{d}t, \qquad (-1)^n \widetilde{f}^{(n)}(a) = \int_0^1 t^n e^{-at}\,\mathrm{d}t > 0.$$

Nevertheless, writing $d = a - b \neq 0$, the off-diagonal divided difference is

$$\mathcal{K}_{\widetilde{g}}(a,b) = e^{-(a+b)/2}\frac{\sinh(d/2)}{d/2}.$$

The determinant of the corresponding two-by-two matrix equals

$$e^{-a-b}\left[1 - \left(\frac{\sinh(d/2)}{d/2}\right)^2\right] < 0. \tag{4.7}$$

The strict inequality follows from $|\sinh x| > |x|$ for nonzero real x, which follows by integrating $\cosh t > 1$ away from zero. Thus these scalar monotonicity conditions cannot be substituted for the fixed-resolvent structure.

5 Holomorphic extension of the model

Choose a real number $M > \|K\|$ with $M > 0$, and set

$$D_M = \mathbb{C} \setminus (-\infty, -1/M]. \tag{5.1}$$

This open connected domain contains zero and the entire positive real axis. The bounded self-adjoint spectral theorem puts $\sigma(K)$ in $[0, M]$ [6]. For $z \in D_M$ and $t \in [0, M]$, the number $1 + zt$ is nonzero. Indeed, a zero would require $t > 0$ and $z = -1/t \leq -1/M$. The functional calculus therefore defines

$$R_z = (I + zK)^{-1}, \qquad F(z) = \langle v, R_z v\rangle \qquad (z \in D_M). \tag{5.2}$$

Lemma 5.1 (Norm-holomorphic resolvent)**.** *The maps $z \mapsto R_z$ and $z \mapsto F(z)$ are holomorphic on D_M.*

Proof. Fix $z_0 \in D_M$. For sufficiently small h, $\|hKR_{z_0}\| < 1$ and

$$I + (z_0 + h)K = (I + z_0K)(I + hKR_{z_0}),$$
$$R_{z_0+h} = R_{z_0} \sum_{n=0}^{\infty} (-hKR_{z_0})^n.$$

The series converges in operator norm on a neighborhood of $h = 0$, so it is a norm-convergent local power series. The scalar functional $T \mapsto \langle v, Tv\rangle$ is continuous and complex linear under the convention in Definition 1.1, and applying it proves the scalar assertion. □

It is useful to retain zero in D_M. A cut at $(-\infty, 0]$ would still suffice for the normalized seed-to-RH argument, but it would hide the evaluation at zero that removes the integration constant in the differential seed. Boundedness of K supplies a genuine neighborhood of zero on which the resolvent is analytic.

The spectral theorem also gives a scalar measure. If E_K is the projection-valued spectral measure, define

$$\mu(B) = \langle v, E_K(B)v\rangle \qquad (B \text{ Borel in } [0, M]). \tag{5.3}$$

It is a finite positive measure, with total mass $\|v\|^2$, and

$$F(z) = \int_{[0,M]} \frac{1}{1 + zt}\, \mathrm{d}\mu(t). \tag{5.4}$$

For a compact $C \subset D_M$, the continuous function $|1 + zt|$ has a strictly positive minimum on $C \times [0, M]$. This bounds every fixed derivative of the integrand by an integrable constant. Thus

$$F^{(n)}(z) = (-1)^n n! \int_{[0,M]} \frac{t^n}{(1 + zt)^{n+1}}\, \mathrm{d}\mu(t). \tag{5.5}$$

This justifies the integral differentiations directly and independently checks the operator formulas of Section 3.

6 The analytic transfer to RH

6.1 Exact calibration on an interval is enough

Theorem 6.1 (Normalized seed implies RH)**.** *If HSeed_Θ holds, then* RH *holds. For its fixed seed data, $F(a) = L(a)$ for all $a \geq 0$.*

Proof. On the interval in Definition 1.1, the calibration and $\Theta(a) > 0$ imply

$$\Theta'(a) - F(a)\Theta(a) = 0.$$

The function

$$G(z) = \Theta'(z) - F(z)\Theta(z) \tag{6.1}$$

is holomorphic on D_M. Its zeros have an accumulation point there, so the identity theorem gives

$$\Theta'(z) = F(z)\Theta(z) \qquad (z \in D_M). \tag{6.2}$$

This passage does not assume that Θ is zero-free on D_M. The expression (6.1) has no division by Θ.

Suppose, for a contradiction, that Θ has a zero $z_0 \in D_M$. Since $\Theta(0) > 0$, it is not identically zero. Its zero at z_0 has some finite order $m \geq 1$, so near z_0,

$$\Theta(z) = (z - z_0)^m h(z), \qquad h(z_0) \neq 0,$$

where h is holomorphic. Substituting in (6.2) and cancelling $(z - z_0)^{m-1}$ off z_0 gives

$$mh(z) + (z - z_0)h'(z) = (z - z_0)F(z)h(z).$$

Both sides are continuous at z_0; evaluation there yields $mh(z_0) = 0$, a contradiction. Therefore every zero of Θ lies outside D_M, hence in $(-\infty, -1/M] \subset (-\infty, 0)$. Proposition 2.1 gives RH. Finally, restricting (6.2) to $a \geq 0$ and dividing by the already known positive $\Theta(a)$ proves $F(a) = L(a)$ everywhere on that ray. □

The proof uses neither a preselected logarithm of Θ on the cut plane nor an RH-dependent product. Choosing a global logarithm before proving zero-freeness would conceal the central difficulty. The division-free ODE avoids that issue.

6.2 The integration constant is real and must be treated

Suppose only the differential identity (1.5) holds. From (3.8),

$$\frac{\mathrm{d}}{\mathrm{d}a}(aL(a) - af(a)) = 0 \quad (a \in (\alpha, \beta)).$$

The ordinary mean value theorem on subintervals shows that

$$a(L(a) - F(a)) = C, \qquad L(a) = F(a) + \frac{C}{a} \tag{6.3}$$

for a constant $C \in \mathbb{R}$. Neither the derivative identity nor the off-diagonal Gram identity alone detects this constant: adding C to $aL(a)$ changes neither its derivative nor its divided differences.

For an arbitrary function on an interval bounded away from zero, one would need a normalization value to set $C = 0$. The actual entire Θ supplies a stronger analytic normalization.

Theorem 6.2 (Differential seed and normalized seed agree)**.** *For the actual Θ, $\mathsf{DSeed}_\Theta \iff \mathsf{HSeed}_\Theta$. The same operator and vector can be used in both directions.*

Proof. If HSeed_Θ holds, differentiating $aL(a) = af(a)$ and using (3.8) gives DSeed_Θ. Conversely assume DSeed_Θ and let C be the constant in (6.3). Consider

$$Q(z) = z\Theta'(z) - (zF(z) + C)\Theta(z). \tag{6.4}$$

This is holomorphic on all of D_M, including zero. By (6.3), it vanishes on (α, β), so the identity theorem gives $Q = 0$ throughout D_M. At zero,

$$0 = Q(0) = -C\Theta(0).$$

Equation (2.5) therefore forces $C = 0$. It follows that $z(\Theta' - F\Theta) = 0$ on D_M. Away from zero we may divide by z, and continuity supplies the equality at zero. Thus (6.2) holds, and positive-ray division by $\Theta(a)$ gives the normalized calibration. In particular HSeed_Θ holds. □

Corollary 6.3 (The recovered norm-square identity is an RH criterion)**.** *For a fixed bounded nonnegative self-adjoint K and fixed v, the identity (1.5) on any nonempty positive interval implies RH. No extra origin-normalization premise is needed beyond the established facts about the actual Θ.*

The qualifier "actual" is essential. If one replaced Θ by $z^m\Theta$ with $m > 0$, its logarithmic derivative would acquire m/z, and the derivative of aL would not see that origin factor. The known value $\Theta(0) \neq 0$ is precisely what rules out this ambiguity here.

7 The converse: constructing a seed under RH

7.1 Why the product has no exponential ambiguity

We use two classical facts about entire functions. The completed ξ has order at most 1; a direct growth argument from the theta integral is given in [4]. Hadamard factorization and its zero-summability consequence are stated in [5, Theorems 3.7 and 3.11]. To see exactly how these apply, write $M_E(r) = \max_{|w|=r} |E(w)|$ and define M_Θ similarly. The map $w \mapsto w^2$ maps the circle of radius $\sqrt{r}$ onto the circle of radius r, so

$$M_\Theta(r) = M_E(\sqrt{r}). \tag{7.1}$$

A bound $|E(w)| \le C_\varepsilon \exp(C_\varepsilon |w|^{1+\varepsilon})$ therefore puts the order of Θ at most $1/2$. Only the strict inequality "order less than 1" is needed below.

For an entire function of order less than 1, nonzero at zero, Hadamard factorization has genus zero and a constant exponential factor. Its nonzero zeros z_n, repeated by multiplicity, satisfy $\sum_n |z_n|^{-1} < \infty$. Hence

$$\Theta(z) = \Theta(0) \prod_n \left(1 - \frac{z}{z_n}\right). \tag{7.2}$$

There is no factor e^{bz} with $b \neq 0$: the polynomial in the Hadamard exponential has degree at most the order, hence degree zero. There is also no origin monomial because $\Theta(0) \neq 0$.

7.2 Zero pairs and multiplicity

Assume RH. Enumerate the positive ordinates of the nontrivial zeros as $\gamma_1, \gamma_2, \ldots$, repeating each according to its multiplicity. There are infinitely many zeros, with no finite accumulation point [3]. We can arrange $0 < \gamma_1 \le \gamma_2 \le \cdots$ with $\gamma_n \to \infty$. The two points $w = \pm i\gamma$ are the preimages of $z = -\gamma^2$. Since the derivative of $w \mapsto w^2$ is nonzero at either preimage, a zero of multiplicity m at either point corresponds to a zero of multiplicity m of Θ. It does *not* become a zero of multiplicity $2m$. Equation (7.2) becomes

$$\Theta(z) = \Theta(0) \prod_{n=1}^{\infty} \left(1 + \frac{z}{\gamma_n^2}\right), \qquad \sum_{n=1}^{\infty} \frac{1}{\gamma_n^2} < \infty. \tag{7.3}$$

Thus one positive/negative pair contributes one factor for each occurrence of its multiplicity.

7.3 Logarithmic differentiation with convergence controlled

Fix a compact set disjoint from the negative zero set, and enclose it in $|z| \le r$. For all sufficiently large n, $\gamma_n^2 > 2r$ and

$$\left|\frac{1}{z + \gamma_n^2}\right| \le \frac{2}{\gamma_n^2}.$$

The right side is summable by (7.3). Finite initial factors have no zeros on the chosen compact set. Locally uniform convergence of the product and its logarithmic derivatives therefore justifies

$$\frac{\Theta'(z)}{\Theta(z)} = \sum_{n=1}^{\infty} \frac{1}{z + \gamma_n^2} \tag{7.4}$$

away from the zeros. In particular this identity holds at every $a \ge 0$.

Proposition 7.1 (Diagonal realization under RH)**.** *Let $\mathcal{H} = \ell^2(\mathbb{N}; \mathbb{C})$, and define*

$$(Kx)_n = \gamma_n^{-2} x_n, \qquad v_n = \gamma_n^{-1}. \tag{7.5}$$

Then K is bounded, self-adjoint, nonnegative, compact and trace class; $v \in \mathcal{H}$; and

$$\langle v, (I + aK)^{-1} v \rangle = L(a) \qquad (a \geq 0). \tag{7.6}$$

Proof. The positive ordinates have a positive minimum: a sequence tending to zero would give an accumulation of zeros at $1/2$, contradicting $\xi(1/2) > 0$. Hence the diagonal entries of K are bounded. They are positive real numbers and tend to zero, which gives self-adjointness, positivity and compactness. Their sum is finite by (7.3), so the positive diagonal operator is trace class. The same sum equals $\sum_n |v_n|^2$, proving $v \in \mathcal{H}$. Finally,

$$\langle v, (I + aK)^{-1} v \rangle = \sum_n \frac{\gamma_n^{-2}}{1 + a\gamma_n^{-2}} = \sum_n \frac{1}{a + \gamma_n^2} = L(a),$$

using (7.4). □

This proves RH $\Rightarrow \mathsf{HSeed}_\Theta$, completing Theorem 1.3. Taking $a = 0$ gives the normalization

$$\operatorname{Tr} K = \|v\|^2 = \sum_n \gamma_n^{-2} = L(0) = \frac{\xi''(1/2)}{2\xi(1/2)}. \tag{7.7}$$

For an arbitrary calibrated representation, $\|v\|^2 = L(0)$ still holds. The equality with $\operatorname{Tr} K$ is asserted only for the particular diagonal construction just given; arbitrary invisible operator components need not have that trace.

If one prefers distinct positive ordinates γ_j with multiplicities m_j, the equivalent scalar model uses $K_{jj} = \gamma_j^{-2}$ and $v_j = \sqrt{m_j}/\gamma_j$. Neither version assumes that zeta zeros are simple. In the repeated model, the vector need not be cyclic when an eigenvalue is repeated. No cyclicity assumption is needed for the argument.

8 Spectral measures, moments and exact reconstruction

8.1 A scalar measure formulation

Define MSeed_Θ to mean that for some $M > 0$ and finite positive Borel measure μ on $[0, M]$,

$$L(a) = \int_{[0,M]} \frac{1}{1 + at} \, \mathrm{d}\mu(t) \tag{8.1}$$

on a nonempty positive interval. Section 5 proves $\mathsf{HSeed}_\Theta \Rightarrow \mathsf{MSeed}_\Theta$. Conversely, given (8.1), take

$$\mathcal{H} = L^2([0, M], \mu), \qquad (Ku)(t) = tu(t), \qquad v(t) = 1.$$

Finite mass makes v square integrable, while multiplication by t is bounded, nonnegative and self-adjoint. Its scalar resolvent is exactly (8.1). Therefore

$$\mathsf{MSeed}_\Theta \iff \mathsf{HSeed}_\Theta \iff \text{RH}. \tag{8.2}$$

The word "finite" in *finite measure* means finite total mass. It does not mean finite support, a finite list of rational numbers, or a finite-checker witness. Under RH the measure is

$$\mu = \sum_n \gamma_n^{-2} \delta_{\gamma_n^{-2}}, \qquad \mu([0, M]) = \kappa. \tag{8.3}$$

This differs from the zero-counting measure $\sum_n \delta_{\gamma_n^2}$, which has infinite total mass. The reciprocal variable and the weights are essential to the normalization.

8.2 Moments and positive finite sections

For any fixed positive operator model, put

$$m_n = \langle v, K^n v\rangle = \int_{[0,M]} t^n \, \mathrm{d}\mu(t) \qquad (n \geq 0). \tag{8.4}$$

If $|z|M < 1$, the Neumann series converges uniformly in the spectral variable and gives

$$F(z) = \sum_{n=0}^{\infty} (-1)^n m_n z^n, \qquad 0 \leq m_n \leq M^n m_0. \tag{8.5}$$

The remainder after degree N has the exact form

$$F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n = (-z)^{N+1} \int \frac{t^{N+1}}{1+zt} \, \mathrm{d}\mu(t), \tag{8.6}$$

and consequently

$$\left| F(z) - \sum_{n=0}^{N} (-1)^n m_n z^n \right| \leq \frac{m_0 (M|z|)^{N+1}}{1 - M|z|}. \tag{8.7}$$

For a calibrated model,

$$m_n = \frac{(-1)^n}{n!} L^{(n)}(0). \tag{8.8}$$

This is an all-orders identity for actual derivatives.

For a polynomial $p(t) = \sum_i c_i t^i$, positivity of μ gives

$$\sum_{i,j} \overline{c_i} c_j m_{i+j} \geq 0, \tag{8.9}$$

$$\sum_{i,j} \overline{c_i} c_j m_{i+j+1} \geq 0, \tag{8.10}$$

$$\sum_{i,j} \overline{c_i} c_j (M^2 m_{i+j} - m_{i+j+2}) \geq 0. \tag{8.11}$$

These are the integrals of $|p|^2$, $t|p|^2$ and $(M^2 - t^2)|p|^2$. Every finite matrix in this hierarchy is positive semidefinite. The following converse records exactly what an *all-orders* moment construction can accomplish.

Theorem 8.1 (Bounded positive moment reconstruction)**.** *Let $(m_n)_{n\geq 0}$ be a real sequence and $M > 0$. Suppose (8.9)–(8.11) hold for every finite coefficient list. Then there exist a Hilbert space, a bounded nonnegative self-adjoint operator K with $\|K\| \leq M$, and a vector v such that $m_n = \langle v, K^n v\rangle$ for all n.*

Proof. Define a complex linear functional ℓ on polynomials by $\ell(t^n) = m_n$, and a sesquilinear form by $\langle p, q\rangle_0 = \ell(\overline{p}\, q)$, where conjugation acts on coefficients and t is real. Condition (8.9) makes this form positive semidefinite, hence it satisfies Cauchy–Schwarz. Quotient the polynomial space by the vectors of zero seminorm and complete to a Hilbert space $\mathcal{H}$.

Multiplication by t is well defined on the quotient: by (8.11),

$$\|tp\|_0^2 = \ell(t^2|p|^2) \leq M^2 \ell(|p|^2) = M^2 \|p\|_0^2.$$

Thus it sends every null vector to a null vector and is bounded by M. It extends to an everywhere defined bounded operator K on $\mathcal{H}$. Since $\ell(\overline{tp}\, q) = \ell(\overline{p}\, tq)$, it is symmetric on a dense subspace and hence self-adjoint after extension. Condition (8.10) gives $\langle [p], K[p]\rangle = \ell(t|p|^2) \geq 0$; continuity makes K positive on all of $\mathcal{H}$. Let $v = [1]$. Then $K^n v = [t^n]$ and $\langle v, K^n v\rangle = \ell(t^n) = m_n$, as required. □

Corollary 8.2 (An all-order actual-jet criterion)**.** *Define m_n by (8.8). RH is equivalent to the existence of $M > 0$ such that (8.9)–(8.11) hold for every finite coefficient list.*

Proof. Under RH, the diagonal seed supplies these inequalities. Conversely, Theorem 8.1 constructs K, v with the specified moments. The resolvent has Taylor series (8.5). The actual L is analytic near zero and has exactly the same Taylor coefficients. They therefore agree on a common disk of positive radius, and in particular on a positive open interval. This is HSeed_Θ, which implies RH. □

The quantifier "for every finite coefficient list" ranges over unbounded degrees. Checking finitely many matrices does not establish this corollary. A finite certificate would have to derive the entire hierarchy by an independently proved uniform mechanism.

8.3 What is determined by calibration

If two compactly supported finite positive measures represent the same actual L on an interval, analytic continuation gives equality near zero. Their moments consequently agree. View both on a common compact interval containing their supports. Polynomials are dense in the continuous functions on that interval, and equality of integrals for polynomials extends, by the total-mass bound, to all continuous functions. Uniqueness of finite Borel measures then gives equality of the two measures. Thus the scalar spectral measure seen by v is determined. This does not identify every operator on the ambient Hilbert space: an orthogonal summand invisible to v leaves all the scalar identities unchanged.

9 Basepoint shifts and finite truncations

9.1 Two different shifted vectors

Fix $a_0 \geq 0$, and abbreviate $R_0 = (I + a_0K)^{-1}$ in this subsection only. Set

$$B = KR_0, \qquad w = R_0^{1/2}v, \qquad u = R_0v. \tag{9.1}$$

These two vectors have different purposes. For $t = a - a_0$ with $a \geq 0$,

$$I + aK = (I + a_0K)(I + tB).$$

All the displayed operators commute. Therefore

$$f(a_0 + t) = \langle w, (I + tB)^{-1}w\rangle. \tag{9.2}$$

The functional calculus gives $0 \leq B \leq M(1 + a_0M)^{-1}I$ whenever $0 \leq K \leq MI$. In particular the local shifted inverse is analytic at $t = 0$.

On the other hand, (3.4) gives

$$g(a_0 + t) - g(a_0) = t\langle u, (I + tB)^{-1}u\rangle. \tag{9.3}$$

To verify this, write $aR_a - a_0R_0 = tR_aR_0$ and $R_a = (I + tB)^{-1}R_0$, then take the scalar product with v. Differentiating (9.3) gives

$$g'(a_0 + t) = \|(I + tB)^{-1}u\|^2. \tag{9.4}$$

The additive value $g(a_0)$ and the distinction $w \neq u$ in general must be retained. Merely replacing a by $a - a_0$ in part of a formula changes the mathematical condition.

9.2 What finite diagonal truncations do prove

Under RH, use the repeated-ordinate model from Proposition 7.1, and truncate after N entries:

$$F_N(z) = \sum_{n=1}^{N} \frac{1}{z+\gamma_n^2}, \qquad S_N = \sum_{n>N} \gamma_n^{-2}.$$

For every $a \geq 0$,

$$0 \leq L(a) - F_N(a) \leq S_N. \tag{9.5}$$

To state a bound on a whole disk, define the tail directly by

$$T_N(z) = \sum_{n>N} \frac{1}{z+\gamma_n^2}.$$

If $\gamma_{N+1}^2 > 2r$, this series is holomorphic on a neighborhood of $|z| \leq r$. Away from the initial poles it equals $F - F_N$; at a shared pole among the first N terms, it is the holomorphic extension of that difference after cancellation. Thus, on the entire disk,

$$|T_N(z)| \leq 2S_N. \tag{9.6}$$

For each derivative order $j \geq 0$, the same argument gives

$$|T_N^{(j)}(z)| \leq j!\, 2^{j+1} \sum_{n>N} \gamma_n^{-2j-2}. \tag{9.7}$$

These estimates follow term by term from the convergent series and its displayed denominator bounds. They show what a tail certificate would need to control, including its dependence on derivative order.

The construction here assumes RH to identify the positive spectrum. It is consequently not an unconditional finite computation proving RH. Even a uniform small error does not imply exact equality; a uniform error bound for values alone does not automatically supply an error bound for all derivatives on the same domain.

10 Why a finite matrix is not the finite witness

10.1 An obstruction to exact finite-dimensional calibration

Proposition 10.1 (Finite visible spectrum is impossible)**.** *There is no exactly calibrated seed for the actual Θ whose scalar spectral measure has finite support. In particular, no finite-dimensional or finite-rank operator can supply an exact seed.*

Proof. A finitely supported spectral measure makes $F(z) = \sum_{j=1}^{q} c_j/(1+\lambda_j z)$ rational, with a possible constant term when $\lambda_j = 0$. Write $F = p/q_0$ for polynomials, where q_0 is not identically zero. Exact interval calibration implies that the entire function

$$q_0(z)\Theta'(z) - p(z)\Theta(z)$$

vanishes on an interval and therefore vanishes on all of $\mathbb{C}$. At any zero z_0 of Θ with $q_0(z_0) \neq 0$, the local ODE $\Theta' = (p/q_0)\Theta$ contradicts the finite-order zero argument of Theorem 6.1. Thus every zero of Θ would have to be among the finitely many roots of q_0. But ξ has infinitely many distinct zeros, and the two-to-one square map can identify only opposite pairs. Hence Θ has infinitely many distinct zeros, a contradiction.

If K is finite rank, its nonzero spectral values form a finite set; its kernel can contribute only the additional spectral point zero. Its scalar spectral measure is therefore finitely supported, so the same argument applies. □

This obstruction does not prohibit a finite description of an infinite-dimensional operator or a finite certificate for an infinite analytic argument. It rules out the narrower proposal that a finite positive matrix by itself can reproduce the full actual logarithmic derivative exactly on an interval.

10.2 The two missing checker theorems

The project's requested finite endpoint is

$$(\exists W : \mathsf{FiniteWitness},\ \mathsf{check}(W) = \mathsf{true}) \quad \Longleftrightarrow \quad \mathrm{RH}. \tag{10.1}$$

The witness must consist of finite raw data and the checker must terminate. The analytic seed defined in this paper is not such a witness type: it quantifies over a Hilbert space, an operator, a vector, and an exact interval identity.

A route through Theorem 1.3 would require the following separate statements for one specified checker:

$$\mathsf{check}(W) = \mathsf{true} \implies \mathsf{HSeed}_{\Theta}, \qquad \text{for every finite } W; \tag{10.2}$$

$$\mathrm{RH} \implies \exists W,\ \mathsf{check}(W) = \mathsf{true}. \tag{10.3}$$

Equation (10.2) is actual-function soundness. It must derive calibration, not receive an unverified calibration proof as a field of the purported raw witness. Equation (10.3) is finite-language completeness. Constructing a countable spectrum from all zeros under RH proves an analytic existence statement; it does not extract an accepted finite code in an independently fixed language.

Finite moment matrices, finite Gram matrices and finite truncations are legitimate components of a possible certificate design. To yield (10.2), they need a theorem extending the verified finite information to the uniform analytic assertion. The all-order moment criterion in Corollary 8.2 specifies one precise target for such an extension. No such finite reduction is established by the present note.

10.3 Finite jets cannot replace a functional identity

There is a direct mathematical reason for the distinction. Fix a real anchor $a_* > 0$ and an integer $N \geq 0$. The functions

$$L(z) \quad \text{and} \quad L(z) + \varepsilon(z - a_*)^{N+1}$$

have identical derivatives through order N at a_* for every ε, but are different analytic functions when $\varepsilon \neq 0$. Finitely many samples can likewise be preserved by adding a nonzero polynomial vanishing at all sample points. The identity theorem requires a set with an accumulation point in the analytic domain, or equivalent exact all-order germ information. A finite initial jet is neither of these.

11 Connection with the repository and proof obligations

This note was prepared against the public repository `randogelist/Riemann-Hypothesis-Final` at the source revision

`362d9f8aeaaf9479cc27ceb6b63bdb7456267990`.

The inspected source records an analytic slit-plane admission [7]:

$$\exists A \text{ holomorphic on } \mathbb{C} \setminus (-\infty, 0], \qquad \Theta' + A\Theta = 0 \text{ there.} \tag{11.1}$$

The source declaration SoloboloAdmission.slitODEAdmission_iff_RH has literal target `_root_.RiemannHypothesis`. The resolvent transfer in this paper supplies the coefficient $A = -F$ on that domain, because the smaller slit plane is contained in D_M. The sign is negative: the paper writes $\Theta' = F\Theta$, whereas the source admission writes $\Theta' + A\Theta = 0$.

The source zero-geometry declaration AngularMaster.native_RH_iff_theta_zeros_nonpositive matches Proposition 2.1, allowing nonpositive real zeros before origin nonvanishing is supplied [8]. The repository's positive-product assessment uses the normalization $\sum_n \lambda_n/(1 + a\lambda_n)$ with $\lambda_n = \gamma_n^{-2}$, consistent with (7.6) [9].

The particular legacy archive name RH_SOLOBOLO_SEED_v0_1_0.zip and the named `NativeOperatorSeed` sources were not recovered in the inspected repository tree or targeted searches. Accordingly, Definitions 1.1 and 1.2 are explicit mathematical formulations supplied by this paper. No claim is made that their fields are textually identical to an unrecovered legacy declaration.

Layer	Statement or construction	Evidence in this note
Resolvent algebra	Norm-square identity and all-parameter Gram formula	On-paper proofs, Sections 3–4
Analytic transfer	Exact normalized or differential seed implies actual RH	On-paper proofs, Sections 5–6
Converse	RH gives the positive countable diagonal seed	On-paper proof from classical factorization, Section 7
Moment route	All-order positivity with one support bound gives a seed	On-paper reconstruction, Section 8
Legacy seed	Exact `SeedOn`/`HasSeed` implementation	Not recovered or certified here
Lean verification	Compilation and exact endpoint axiom audit	Not run; no new formal certification
Finite checker	Actual soundness, finite completeness, accepted witness	Open; not constructed here

The repository's inspected status records native build and axiom audit as `NOT_RUN`, actual calibration as `OPEN`, and finite-witness soundness and completeness as `OPEN` [10]. Publication of this paper does not change those statuses. A later formalization must connect the exact definitions, complex derivative, domains, spectral constructions and zero geometry to the actual native declarations; familiar theorem names are not a substitute for that transport.

12 Conclusion

The recovered Hilbert-vector anchor has a complete and coherent on-paper analytic formulation. Its central scalar identity is

$$\frac{\mathrm{d}}{\mathrm{d}a}\left(a\frac{\Theta'(a)}{\Theta(a)}\right) = \|(I + aK)^{-1}v\|^2,$$

where K and v are fixed, K is bounded and nonnegative self-adjoint, and the identity is exact on a positive interval. The proof proceeds through a holomorphic resolvent, an explicitly removed

integration constant, and a division-free analytic ODE. The zero-order contradiction then yields the critical-line statement.

Conversely RH supplies a diagonal seed through the genus-zero product of the entire square-descended function. This gives the analytic equivalence (1.6), the Gram identity, and the normalization (7.7). It does not construct the seed without RH. The unresolved constructive task is to derive one of these exact analytic inputs from independently verifiable data; the further finite-checker equivalence requires both (10.2) and (10.3).

A Real Hilbert spaces and the holomorphic convention

The recovered anchor may be expressed on a real Hilbert space. That does not obstruct the complex-analytic proof. If H is real, form $H_{\mathbb{C}} = H \oplus iH$ with

$$\begin{aligned}\langle x+iy, u+iw\rangle_{\mathbb{C}} = (x,u)_H + (y,w)_H \\ + i\big((x,w)_H - (y,u)_H\big).\end{aligned}$$

This inner product is conjugate-linear in the first slot and linear in the second. A real bounded self-adjoint nonnegative K extends by $K_{\mathbb{C}}(x+iy) = Kx + iKy$. Its norm is unchanged, because

$$\|K_{\mathbb{C}}(x+iy)\|^2 = \|Kx\|^2 + \|Ky\|^2 \le \|K\|^2(\|x\|^2 + \|y\|^2),$$

and restriction to real vectors gives the reverse norm inequality. Self-adjointness follows by expanding the inner product. Positivity follows from

$$\langle x+iy, K_{\mathbb{C}}(x+iy)\rangle = (x,Kx)_H + (y,Ky)_H \ge 0,$$

since the imaginary cross terms cancel by symmetry. For real $a \ge 0$, the inverse extends the real inverse, so every real scalar calibration and norm identity is preserved. Thus the complex theorem applies to the real Hilbert version as well.

If an alternative convention makes the inner product linear in the first slot, the scalar holomorphic expression should instead be $\langle R_z v, v\rangle$. With the convention fixed in this paper, it is $\langle v, R_z v\rangle$. Mixing these conventions would turn a holomorphic function into an anti-holomorphic one and invalidate the identity-theorem step.

B A compact proof dependency ledger

The forward implication has the following dependencies, each proved or cited at its point of use:

- **F1.** The actual Θ is entire and $\Theta(0) > 0$.
- **F2.** Fixed bounded positive K gives a holomorphic resolvent on D_M, a connected domain containing zero and the positive ray.
- **F3.** Exact interval calibration gives $\Theta' = F\Theta$ by the identity theorem. For differential calibration, first remove C using $Q(0) = -C\Theta(0)$.
- **F4.** A nonzero entire solution of that ODE has no zero where its coefficient is holomorphic, by the finite-order contradiction.
- **F5.** Negative-real-axis confinement of Θ zeros is equivalent to the critical-line condition for actual ξ zeros.

Only the converse uses RH:

- **R1.** RH places the zeros of Θ at $-\gamma_n^2$.

R2. The order bound and Hadamard factorization give $\Theta/\Theta(0) = \prod_n (1 + z/\gamma_n^2)$ and summability.

R3. Uniformly convergent logarithmic differentiation gives $L(a) = \sum_n (a + \gamma_n^2)^{-1}$.

R4. The explicit diagonal K and square-summable v realize this scalar function and hence both seed conditions.

The all-order moment criterion reconstructs the forward input through Theorem 8.1. It does not remove an infinite quantifier by itself. Neither direction in this ledger uses a finite accepted witness, and neither is reported as a Lean-verified endpoint.

References

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[6] Dana P. Williams, *Lecture Notes on the Spectral Theorem*, 10 August 2018, Sections 2, 4 and 5 (bounded spectral calculus and spectral measures). https://math.dartmouth.edu/~dana/bookspapers/ln-spec-thm.pdf.

[7] Riemann-Hypothesis-Final project, `SoloboloAdmission/NativeODE.lean`, source revision `362d9f8`. Exact source of `slitODEAdmission_iff_RH`; native audit not run in this note. Pinned repository source.

[8] Riemann-Hypothesis-Final project, `AngularMaster/ZeroGeometry.lean`, source revision `362d9f8`. Pinned repository source.

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[10] Riemann-Hypothesis-Final project, `provenance/status.json`, `docs/BUILD.md`, and `docs/FRONTIER.md`, source revision `362d9f8`. Source evidence and open finite calibration obligations. Pinned status record.